PROCESS DEFINITION DOCUMENT (PDD)

Project 2 • Supply Chain Material Tracker & Dashboarding

Document purpose

This PDD describes the business process and proposed automation for consolidating daily material-status data from multiple sources into one reporting output and refreshing supply-chain visibility through Power BI.

1. Process overview

Item	Description
Process name	Supply Chain Material Tracker & Dashboarding
Business area	Supply Chain / Materials / Operations
Automation platform	Automation Anywhere
Source systems	SAP dumps, web portals, OneDrive and approved source files
Reporting tools	Excel and Power BI
Source volume	5+ disparate sources, as described in the project summary
Reported impact	Eliminated about 3 hours of daily manual effort; reporting lag reduced by 70%

2. Business objective

Automate the daily collection, consolidation and validation of material-status information from approved supply-chain sources. Publish a consistent consolidated output for business users and make the information available through Power BI so teams can identify delays, shortages and status changes faster.

3. Scope

In scope

- Collect daily material-status data from the configured SAP dump files, web portals and OneDrive locations.

- Validate file availability, expected columns, date formats and mandatory identifiers.
- Standardize field names and formats and consolidate source data into one reporting dataset.
- Identify duplicate rows, missing values and source-level exceptions using approved rules.
- Save the consolidated dataset or Excel output to the agreed OneDrive/shared location.
- Refresh or prepare data for the Power BI dashboard according to the approved refresh method.
- Record execution status, source-level counts and exceptions for operational follow-up.

Out of scope

- Changing material master data, purchase orders or inventory records in SAP.
- Making procurement, allocation or supplier decisions on behalf of business owners.
- Modifying source-system configuration or access permissions.
- Changing Power BI semantic models or measures without report-owner approval.
- Automatically correcting ambiguous business data without an approved rule.

4. Stakeholders and responsibilities

Role	Responsibility
Supply chain / materials team	Defines material-status meaning, required fields, business rules and exception priorities.
Source-system owners	Maintains source access, file structure and portal availability.
Power BI / reporting owner	Confirms dataset structure, refresh approach, dashboard measures and access.
RPA developer / support	Builds and supports extraction, transformation, logging and recovery.
Process owner	Approves scope, acceptance criteria, schedule and operational handover.

5. Inputs and prerequisites

- Configured list of source systems, URLs, folders, file patterns and expected arrival times.
- Approved credentials and permissions for each source, stored through the organization's secure mechanism.
- A data dictionary mapping source columns to the common reporting schema.

- Business rules for duplicates, missing data, status mapping, date/time handling and late-arriving sources.
- Approved output folder, file naming convention, retention period and Power BI refresh method.
- A defined owner for source failures and material-status exceptions.

6. Proposed process steps

Step	Action	Expected result
1	Start scheduled/manual bot and load run configuration.	Run ID, run date and source list are recorded.
2	Check access to each configured source.	Available and unavailable sources are identified.
3	Extract/download the latest approved SAP dumps, portal data and OneDrive files.	Raw input files/data are collected without changing source records.
4	Validate file structure and mandatory fields.	Valid data proceeds; invalid inputs are flagged.
5	Normalize column names, date formats, material identifiers and status values.	Sources align to the common data model.
6	Combine source data and apply approved deduplication rules.	One consolidated dataset is created.
7	Run quality checks and compare source/processed record counts.	Quality summary and exceptions are available.
8	Save the output to the approved location.	Versioned consolidated Excel/CSV dataset is available.
9	Refresh or stage data for Power BI using the configured method.	Dashboard reflects the latest successfully processed data.
10	Log outcome and notify owners of completion or exceptions.	Business users know whether the run completed and what needs attention.

7. Business rules and controls

- Use only the approved sources and file patterns; do not silently substitute stale data.
- Retain source timestamps and a run timestamp so data freshness can be assessed.

- Use a business-approved unique key for duplicate detection; do not assume material ID alone is unique unless confirmed.
- Do not convert missing or unknown statuses into a valid status without an approved mapping.
- If a critical source is missing, follow the agreed policy: stop the full run or publish a clearly marked partial dataset.
- Do not overwrite the last known-good output until the new output passes validation, unless an approved process says otherwise.
- Apply least-privilege access and do not expose credentials or unnecessary sensitive data in logs or dashboards.

8. Exceptions and recovery

Exception	Bot response	Owner / next action
Source unavailable / portal timeout	Log source and time; retry only within configured limits.	Source owner / RPA support
Expected file missing	Mark source as missing and apply critical-source policy.	Source owner / supply chain team
Schema or column change	Quarantine affected input and do not map fields by guesswork.	Source owner / RPA developer
Invalid dates or mandatory fields	Flag rejected rows and provide a validation summary.	Business data owner
Duplicate or conflicting status	Apply approved precedence rule or flag for review.	Supply chain process owner
Output save failure	Mark run failed; preserve prior validated output.	RPA / storage support
Power BI refresh failure	Log refresh failure and communicate dashboard freshness status.	Power BI owner / support

9. Outputs and evidence

- Consolidated material-status dataset/report in the approved Excel or other agreed format.
- Power BI dashboard refresh or refresh-ready dataset, depending on configured integration.
- Run log with source availability, timestamps, record counts, status and sanitized error summaries.

- Data-quality summary covering missing fields, duplicates, rejected rows and late sources.
- Completion or exception notification to the agreed stakeholder group.

10. Assumptions and open points

- The exact source names, portal URLs, SAP dump formats and source owners must be confirmed.
- The common data schema, material unique key and source-precedence rules require business approval.
- The method for triggering Power BI refresh (for example, gateway, dataset refresh or staged file) must be confirmed with the report owner.
- The reported 3 hours saved per day and 70% reduction in reporting lag are project-stated outcomes and should be validated against measured baselines.
- Run schedule, late-source cutoff, partial-data policy, notification channel and retention period remain to be finalized.

11. Acceptance criteria

- All configured sources are processed or clearly identified as unavailable.
- Records are mapped into the approved common schema and pass agreed quality checks.
- Duplicate handling and missing-data behavior match approved business rules.
- The consolidated output is saved to the agreed location without unintended overwrites.
- Power BI reflects the latest successful dataset refresh, or the refresh failure is clearly logged.
- Business users validate the output against sample source records and approve the process for production.